

Project ID:

25-26J-157

1. Topic (12 words max)

Decision Support System for Agarwood Cultivation

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering (ISE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Agarwood (*Aquilaria* spp.) is a highly valued resin-producing tree that is used worldwide in perfumes, incense, and traditional medicine. Farmers in Sri Lanka and other Southeast Asian countries are becoming more interested in agarwood growing due to its great economic worth. However, establishing agarwood trees and harvesting excellent resin is a lengthy (usually over 8 years) labor-intensive procedure that relies significantly on manual decision-making and traditional knowledge. This causes substantial hurdles in productivity, efficiency, and consistent yield. The inoculation stage, in which the tree is intentionally damaged or injected to encourage resin production, is the most important step in agarwood production. Currently, most farmers use visual cues such as tree death or color changes to evaluate if a tree is suitable for resin harvest. However, the absence of a consistent assessment process frequently leads to uncertainty, premature harvesting, or misidentification of trees, resulting in financial loss. Furthermore, farmers lack scientific instruments to assess soil quality, tree age, post-inoculation progress, and the best time or location for resin harvesting. These problems highlight the critical need for a decision support system (DSS) customized to the unique lifespan of agarwood trees.

References:

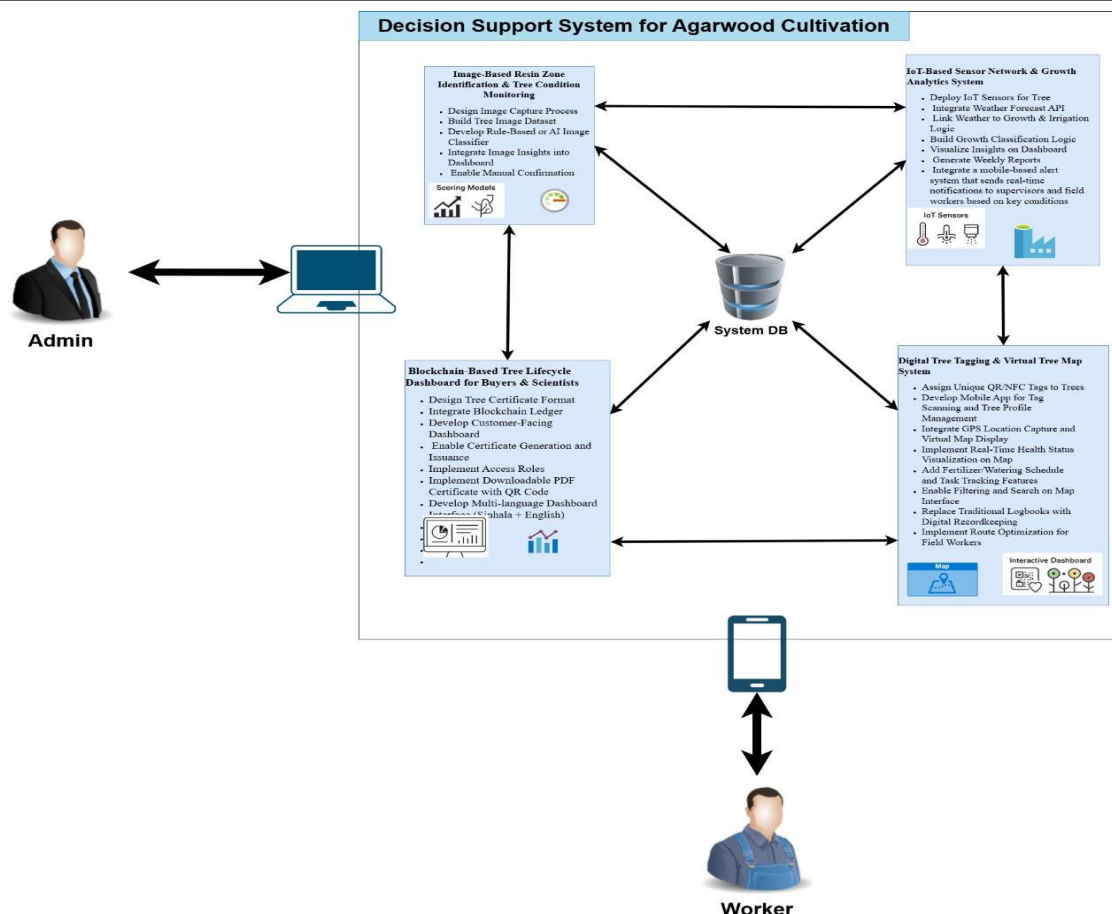
<https://www.sciencedirect.com/science/article/abs/pii/S092666901500062X>
<https://www.sjp.ac.lk/research/walla-patta-gyrinops-walla-for-agarwood-production/>
https://link.springer.com/chapter/10.1007/978-1-4020-8261-0_12
<https://www.mdpi.com/1420-3049/18/3/3086>

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This research suggests the creation of a smart IoT-based decision support system tailored exclusively to agarwood plantations. The system consists of four key components:

- real-time soil and environmental monitoring via IoT sensors
- data patterns and virtual tree mapping with tagging (NFC tags/QR/RFID) for tracking individual tree history.
- prediction of high-resin zones using scoring models
- blockchain-backed certification and traceability dashboard

Each component is intended to address a specific bottleneck in current cultivation processes and promote data-driven decision-making. The project's novelty stems from the integration of precision agricultural technologies into an under-researched, high-value crop such as agarwood. While similar technologies have been used for short-term crops such as vegetables or grains, they are rarely used for long-term, stress-sensitive crops like agarwood. The system also intends to give both plantation-level insights and tree-level knowledge through the use of low-cost IoT sensors, interactive dashboards, and mobile worker interfaces. This research will aim to assist plantation owners, workers, and authorities by increasing productivity, reducing resource waste, and boosting sustainability. This system, which combines smart agriculture, tree lifecycle analytics, and decision support tools, will provide a realistic and scalable solution to Sri Lanka's rising agarwood industry and other tropical regions.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project necessitates competence in Information and Systems Engineering, namely in the integration of IoT systems, data collecting, sensor networks, and software development. Knowledge of embedded systems (e.g., ESP32, Arduino), wireless communication (Wi-Fi, MQTT), and cloud-based data storage is required to collect and handle real-time environmental data from agarwood crops. Building mobile/web interfaces, dashboards, and alert systems requires both front-end and back-end development skills. Additionally, geospatial mapping, database management, and API connectivity (for example, weather forecasting) are required to enable interactive features such as the virtual tree map and decision assistance dashboards.

On the agricultural side, fundamental knowledge of the agarwood culture lifecycle is required, including growth stages, environmental needs, inoculation timing, and resin formation conditions. The system is dependent on precise and timely data inputs like as sensor readings (temperature, soil moisture, humidity, pH), tree-specific data (age, location, inoculation status), and weather data (predicted rainfall, temperature, and wind). Historical and real-time records are utilized to detect abnormalities, characterize growth trends, optimize irrigation, and direct workers' task scheduling. Using data engineering, system design, and applied field expertise, the project develops a feasible and scalable solution for smart plantation management.

8. Objectives and Novelty

Main Objective: To design and develop a smart decision support system for agarwood plantations in Sri Lanka by integrating IoT-based environmental monitoring, image-based resin analysis, GPS-enabled tree mapping, and blockchain-backed certification to enhance operational efficiency, transparency, and resin yield quality.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Dehigolla D. N. IT22123886	To monitor and analyze environmental and biological conditions for each tree using IoT sensors and forecasted weather data, enabling real-time growth status classification, predictive planning, and long-term trend analysis for better decision-making.	1. Deploy IoT Sensors for Tree - Install sensors to monitor: - Temperature - Soil moisture & humidity - Air quality / resin-related gas emissions 2. Integrate Weather Forecast API - Fetch weather data (temperature, humidity, rainfall, wind) using plantation GPS coordinates. 3. Link Weather to Growth & Irrigation Logic - Combine live sensor data with forecast data to:	This component introduces the first integrated use of IoT sensors and real-time weather forecasting in Sri Lankan agarwood plantations . It transforms manual, reactive monitoring into proactive decision-making , using both local sensor data and predictive weather models. By combining growth analytics with forecasted environmental data, the system offers data-driven recommendations for watering and inoculation, reducing risk from poor timing and maximizing resin yield. The data-driven insights generated through this system will enable plantation owners to replace

		○ Optimize watering schedules ○ Predict tree stress levels ○ Adjust inoculation plans (e.g., avoid during high rain/wind) 4. Build Growth Classification Logic • Classify trees based on current and historical growth data. • Highlight growth decline, abnormal stagnation, or healthy trends. 5. Visualize Insights on Dashboard • Per-tree data: current readings, growth curve, weather forecast • Aggregate views: zone health summary, weather alerts, irrigation guidance • Timeline analysis: growth patterns linked to weather and soil 6. Generate Weekly Reports • Recommend actions like watering, fertilizing, or inoculation	guesswork with evidence-based planning, improving resource use, yield prediction, and economic return.
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		- Summarize tree health, environmental trends, and forecast warnings 7. Integrate a mobile-based alert system that sends real-time notifications to supervisors and field workers based on key conditions such as: - Low soil moisture detected by sensors - Sudden temperature drops or extreme weather forecast - Inoculation follow-up reminders - Abnormal tree growth patterns	
Wijesinghe A.G.T - IT22319142	To digitize tree-level tracking using QR or NFC tags and develop a GPS-based virtual tree map that displays the real-time condition, location, and operational status of each tree, including the area where the tree is physically situated	1. Assign Unique QR/NFC Tags to Trees - Tag each tree with a QR code or NFC chip that contains a unique Tree ID. - The tag acts as the tree's digital identity for lifecycle tracking and profile retrieval. - Ensure durability of physical tags for outdoor use (e.g., waterproof, UV-resistant). 2. Develop Mobile App for Tag Scanning and Tree Profile Management	This will be the first known use of tag-to-digital map integration in Sri Lankan agarwood plantations , introducing a significant leap in plantation management. The system combines IoT-based tracking, GIS (Geographic Information Systems) , and real-time task visibility at the individual tree level. Each tree is uniquely tagged using QR or NFC technology and mapped with GPS coordinates, allowing plantation owners and workers to monitor tree conditions, track ongoing operations,

	within the plantation.	• Allow field workers to scan QR/NFC tags using a mobile device. • On scanning, open a tree profile showing key data: species, planting date, inoculation status, treatment history, and health notes. • Enable workers to add/edit logs, update current status, and attach photos in real-time. 3. Integrate GPS Location Capture and Virtual Map Display • On tagging or updating, capture the tree's GPS location and store it in the system. • Display all trees on a digital plantation map using their coordinates. • Use satellite or grid view to visually represent physical plantation layout. 4. Implement Real-Time Health Status Visualization on Map • Assign a color-coded icon to each tree on the map based on its current status:	and plan field activities more efficiently. By unifying identification, location, health status, and operational data into a single platform, the solution provides a comprehensive tool for both monitoring and strategic plantation management.
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		- o Green – Healthy - o Yellow – Inoculated / Processing - o Red – Under Observation - o Black – Dead or Dormant • Allow users to click on each icon to view/edit the tree’s full profile. 5.Add Fertilizer/Watering Schedule and Task Tracking Features • Display watering and fertilizer schedules per tree, with task progress indicators. • Enable task logging such as “watered,” “fertilized,” “pruned,” etc., with timestamps and responsible personnel. • Allow supervisors to assign upcoming tasks via the app. 6.Enable Filtering and Search on Map Interface • Add filters to show trees by status, type, or location (e.g., “Show only Yellow/Inoculated trees in Zone A”).	
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		• Provide search functionality by Tree ID, species, or worker assigned. • Include legends and toggle layers for better map interaction. 7. Replace Traditional Logbooks with Digital Recordkeeping • Store all data such as treatments, growth notes, and conditions digitally in the cloud. • Enable audit trails and version history of tree updates. • Ensure offline sync capability for low-connectivity areas. 8. Implement Route Optimization for Field Workers • Use GPS and map data to auto-suggest the most efficient route for workers to visit multiple trees needing attention. • Allow workers to generate a daily task path (e.g., 10+ trees marked as Red/Yellow). • Reduce field operation time and ensure no tree is skipped during routine care.	
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Amarasooriya R.P.C.A. IT22123640	To detect resin-rich areas and assess tree condition using image-based analysis, enabling better decision-making for harvesting and ongoing tree management through visual inspection and AI-supported classification.	1. Design Image Capture Process • Provide mobile phones or low-cost cameras (helmet/chest-mounted) to field workers. • Capture images of tree trunks, bark, and leaves during key stages (e.g., post-inoculation) 2. Build Tree Image Dataset • Tag images by tree ID, date, and stage (e.g., pre-inoculation, post-inoculation). • Store in a cloud-based system linked to the digital tree profile. 3. Develop a classified analysis or AI Image Classifier • Identify visual signs of: ◦ Resin formation (bark darkening, sap markings) ◦ Tree damage or fungal infection ◦ Stress signs (leaf drop, discoloration) • Use OpenCV or ML models (e.g., TensorFlow Lite) if dataset allows. 4. Integrate Image Insights into Dashboard	This is the first known attempt in Sri Lankan agarwood plantations to apply image-based AI or develop a classified analysis to detect resin formation and monitor tree condition. Traditionally, workers rely solely on visual judgment, which can be inconsistent and subjective. By capturing and analyzing tree trunk images systematically, this component introduces data-supported visual verification of resin readiness and tree stress. Additionally, the system builds a visual timeline of each tree's appearance over time, supporting traceability and quality assurance. It reduces the chances of cutting immature trees, identifies high-resin harvest zones, and introduces a scalable method for tree condition surveillance, especially useful in large plantations.
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		- Show status like: - Resin visible - Suspected damage - Needs inspection - Link to resin zone heatmap or harvesting recommendation list. 5. Enable Manual Confirmation - Allow workers or supervisors to mark tree zones based on AI/image suggestions. - Help train and refine the image detection model.	
Thennakoon T.A.C.S. IT22310996	To provide transparent, tamper-proof access to tree lifecycle data for buyers, scientists, and regulatory bodies by leveraging blockchain technology, enabling certification, traceability, and	1. Design Tree Certificate Format - Define the digital certificate content: tree ID, planting date, inoculation date, growth summary, resin quality indicators, and GPS location. 2. Integrate Blockchain Ledger - Store key lifecycle events (planting, inoculation, harvesting, certification) as blockchain transactions using platforms like Ethereum, Hyperledger, or a lightweight hash verification system.	This is the first known implementation of a blockchain-backed certification and traceability dashboard in the Sri Lankan agarwood industry. Agarwood cultivation and resin extraction involve long-term, high-value assets, but currently lack transparent verification or tamper-proof documentation. By introducing blockchain to secure lifecycle events, this system provides assurance to external stakeholders; including buyers who want proof of origin and maturity, and scientists

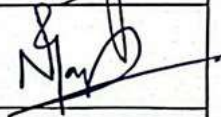
	informed decision-making	3. Develop Customer-Facing Dashboard • Provide a web-based dashboard for buyers or researchers to: ○ View full tree history ○ Verify certificate authenticity ○ See the timeline and status changes ○ Track responsible personnel and resin yield history 4. Enable Certificate Generation and Issuance • Allow verified plantation managers to issue digital certificates post-harvest. • Include a unique blockchain reference or QR code for traceability. 5. Implement Access Roles • Customers: View tree status and certificate • Scientists: Access tree growth data for research • Owners/Admins: Manage blockchain records and approvals	interested in research-backed data. The solution combines smart certification, real-time monitoring, and digital integrity into a single interface, unlocking a new layer of trust and operational efficiency in the plantation value chain.
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		6. Implement Downloadable PDF Certificate with QR Code Generate downloadable PDF certificates for each harvested tree, containing: • Tree ID and history (planting, inoculation, harvesting) • Resin yield and quality indicators • A QR code that links to the blockchain record for authenticity verification 7. Develop Multi-language Dashboard Interface (Sinhala + English) Add a language toggle option to the buyer/scientist dashboard to support both Sinhala and English. Ensure: • All labels, instructions, and content switch language dynamically. • Localized formatting (e.g., dates, numbers) where needed.	
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Dehigolla D. N. IT22123886	This component implements basic Information and Systems Engineering principles by combining IoT sensor networks, weather data APIs, real-time monitoring, and predictive analytics to facilitate intelligent agarwood crop decision-making. System design, data collecting, environmental modeling, and automated logic creation are used to classify tree growth state and optimize field activities. Dashboards, alarm systems, and structured reporting demonstrate the use of system-level thinking, embedded systems, and data-driven insights, which is directly aligned with the specialization's emphasis on developing smart, connected, and efficient systems for real-world challenges.
Wijesinghe A.G.T - IT22319142	This component uses Information and Systems Engineering principles to create a smart identification and tracking system that connects actual trees to a digital interface using QR/NFC tags and GPS-based mapping. It entails creating an integrated system in which real-world data (tree health, location, and task status) is gathered, visualized, and controlled using a centralized platform. The use of interactive tree maps, color-coded condition indicators, real-time updates, and route optimization exemplifies the application of system integration, data visualization, user-centered design, and spatial information processing, all of which are important aspects of the specialization's goal of creating efficient, technology-driven field solutions.
Amarasooriya R.P.C.A. IT22123640	This component is consistent with the Information Systems Engineering specialization since it uses image processing, develop a classified analysis, and AI techniques to solve a real-world agricultural problem. It entails recording, storing, and analyzing tree pictures to aid decision-making, demonstrating important IS skills such as system architecture, data integration, and intelligent decision support. The component also employs cloud storage and dashboard interfaces to show how information systems may increase accuracy, efficiency, and traceability in plantation management.
Thennakoon T.A.C.S. IT22310996	This component complements the Information Systems Engineering specialization by incorporating crucial ideas like blockchain integration, dashboard design, role-based access, and digital certification. It highlights how secure and transparent systems may be built to facilitate data tracing and lifecycle management. The usage of blockchain for tamper-proof event recording and certificate issuing exemplifies advanced IS principles, whereas the customer-facing dashboard, QR verification, and multi-language support demonstrate a strong emphasis on user-centric system development and digital trust, which are central to our IT specialization.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Chathurangika	Kahandawarachchi	
Co-Supervisor	Ms	Narmada	Gamage	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Data & resource intensive project. Require external assistance & supervision.
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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

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Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.